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**Goal (G)**

My graduate goal is to develop advanced expertise in machine learning and natural language processing within the context of Industry 4.0, with a specific focus on smart semiconductor manufacturing systems. I aim to build strong technical depth in applied ML and NLP methods that support process monitoring, decision support, and data-driven optimization in complex manufacturing environments. Through advanced coursework and applied research, I want to strengthen my ability to design, implement, and evaluate models that integrate temporal process data with unstructured textual data, such as maintenance logs, equipment reports, and operational documentation. This goal is central to my doctoral training, as semiconductor manufacturing is a highly data-intensive domain where intelligent analytics can significantly improve efficiency, reliability, and decision-making. Developing this expertise will prepare me for independent research at the intersection of artificial intelligence, engineering systems, and industrial applications.

**Reality (R)**

I am currently making steady progress toward this goal through a combination of coursework and applied projects. I have completed ECE 595 (Natural Language Processing), where I gained hands-on experience with NLP techniques and worked on projects involving real-world text data. In parallel, my coursework in STAT 520 (Time Series Applications) strengthened my ability to model temporal dependencies using real datasets, including ARMA/ARIMA modeling, regression with autocorrelated errors, and diagnostic analysis. These skills are directly relevant to analyzing sensor-driven and process-level data in semiconductor manufacturing.

I am also involved in a project applying NLP methods, including topic modeling, to real-world datasets, which has helped me understand how machine learning models behave outside controlled settings. What is working well is the integration of structured coursework with applied experimentation, allowing me to connect theory with practice. However, challenges remain. Developing and testing advanced ML and NLP models requires significant time for experimentation, parameter tuning, and computational analysis. Balancing coursework, research responsibilities, and experimentation has been difficult at times, limiting how deeply I can explore alternative approaches. Recognizing these constraints has clarified the need for more focused and efficient strategies to fully achieve my technical development goals.

## Options (O)

To move forward, I see several concrete options.

**First,** I can more intentionally align my coursework with my research by directly applying ML, NLP, and time-series methods learned in ECE 595 and STAT 520 to smart semiconductor manufacturing problems, such as analyzing maintenance logs alongside process signals.

**Second,** I can maintain a structured routine for reviewing recent research in Industry 4.0, smart manufacturing, and AI-driven decision support systems, ensuring my work reflects current methods and emerging trends.

**Third,** I can actively collaborate with faculty members and research personnel working in manufacturing systems, AI, or semiconductor-related research, which would allow knowledge sharing, access to relevant data, and more efficient experimentation.